Web Technologies

Responsive Web Design

Media Queries

Summary of Today's Lecture

- Fixed vs Fluid layouts
- Responsive Web Design
- Media Queries
  - Media Types
  - Media Features
  - Logical Operators
- Identifying Breakpoints
- Mobile First Approach
- Adding Media Queries
- Responsive Web Design Example

CSS Layouts
Fixed Vs Fluid

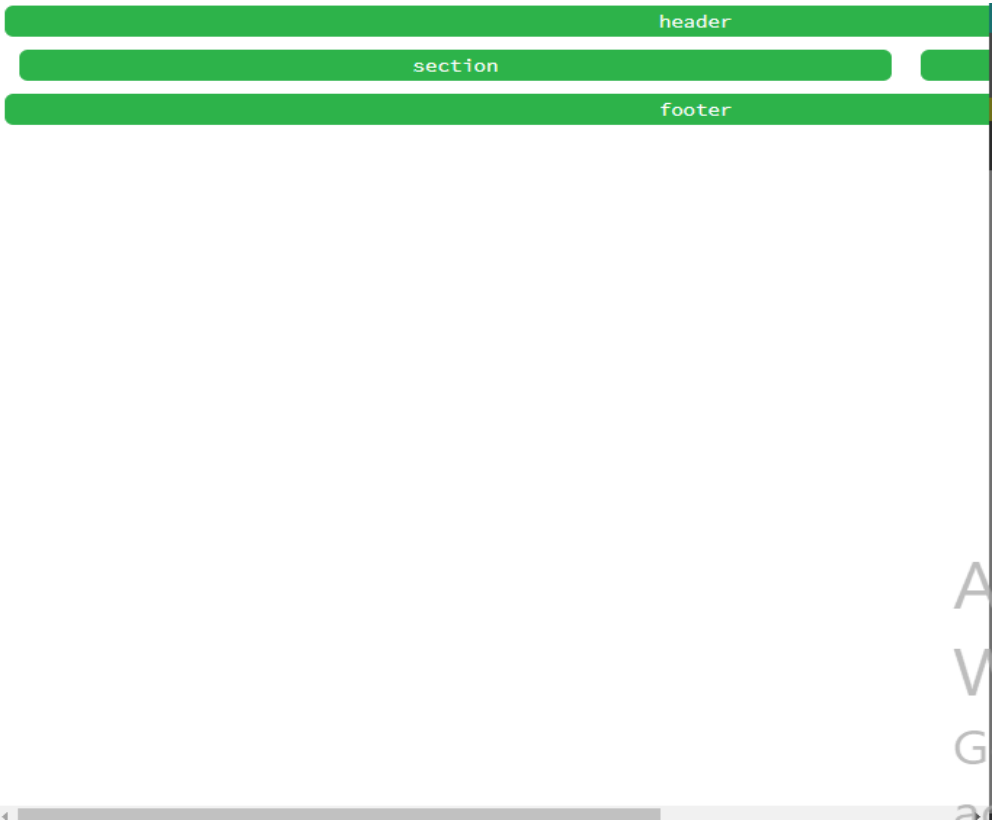
- Till CSS2, we can build two types of layouts, **Fixed** and **Fluid**.
- These layouts are classified as **Fixed or Fluid** on the basis of their width.
  - **CSS Fixed layouts** are layouts with fixed width in px for example, 960px, 1200px, 1170px, etc.
  - **CSS Fluid Layouts** or Elastic layouts are layouts with width in percentage.

Fixed Layout

- Fixed Layout is a layout in which the width of the container is fixed (in pixels).
- Properties of Fixed Layout
  - Fixed width in pixels.
  - Independent of screen size.
  - Horizontal Scroll will come when screen size is less than width of main container.

Fixed Layout Example

```
<main>
    <header> header </header>
    <div id="container">
      <section>section</section>
      <aside> Aside </aside>
    </div>
    <footer>footer</footer>
</main>
```




```
header{width: 950px;}
section {
   float: left;
   width: 600px;
   margin:10px;}

aside {
   float: left;
   width: 300px;
   margin:10px;}

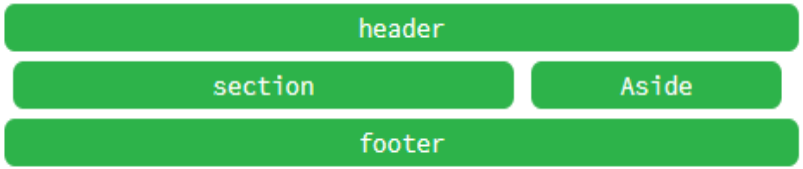
footer {
   clear: left;
   width: 950px;}

main{width:1000px;}
#container{width:1000px;}
```

Fluid Layouts

- Liquid Layouts is a layout in which the width of the container is flexible (in percentage). Whatever the screen-size is, the layout will remain same.
- Flexible layouts is the practice of building the layout of a website with a flexible grid, capable of dynamically resizing to any width. (Ethan)
- Properties of Fluid Layout
  - Width is in percentage.
  - Dependent of screen size.
  - Horizontal Scroll will Never come on any screen unless any fixed content is inside.

Fluid Layout Example




```
header{width: 95%;}
section {
  float: left;
  width: 60%;
  margin:1%;}

aside {
  float: left;
  width: 30%;
  margin:1%;}

footer {
  clear: left;
  width: 95%;}

main{width:100%;}

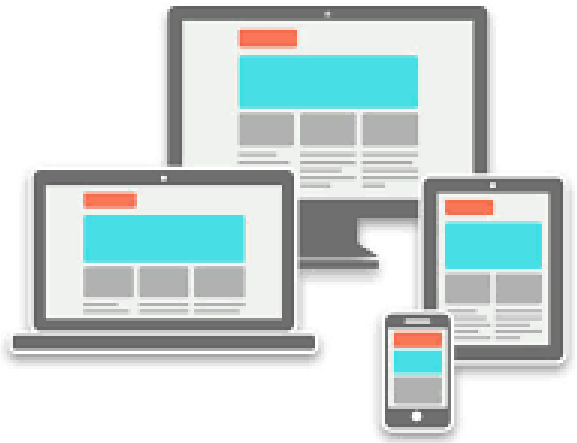
#container{width:100%;}
```

Problem with flexible layout

- The flexible layout approach alone isn’t enough.
- If the width of a browser viewport may be so small that even scaling the layout proportionally will create columns that are too small to effectively display content.
- As a result, when the layout gets too small, text may become unreadable and the layout may begin to break.

Responsive Web Design (RWD)

- Responsive web design is the practice of building a website suitable to work on every device and every screen size, no matter how large or small screes size is.
- Responsive web design is focused around providing an intuitive and gratifying experience for everyone.
- The responsive web design term itself was coined, and largely developed, by `Ethan Marcotte` in his book `Responsive Web Design` in `2010`.
- Responsive web design can be achieved using media queries which is specified as part of CSS3 specification.



Media Queries

- Media queries allow designers to deliver styles based on media type and media features.
  - Media types reflect target devices for which the relevant properties make sense.
  - Media features specify different styles for individual browser and device circumstances, the width of the viewport or device orientation.
- The `@media` rule, introduced in CSS2, made it possible to define different style rules for different media types.
- A media query consists of a media type and can contain one or more media features, which resolve to either `true` or `false`.
  - ```
    @media
    not|only mediatype and (mediafeature and|or|not mediafeature) {
      CSS-Code;
    }
    ```
- For example, following is the media rule that works for all types of screens for the size of screen less than `1024px`.
  - `@media (max-width: 1024px) {...}`
- Generally Media queries are added inside of an existing style sheet.
    ```

Media Types

- Media Types basically define which of the CSS file will be directed for your website based on the type of media, i.e, for each different device that you access a website, you need a different css too.
  - `@media print { ... }`
  - `@media screen, print { ... }`

Value	Description
all	Default. Used for all media type devices
print	Used for printers
screen	Used for computer screens, tablets, smart-phones etc.
speech	Used for screenreaders that "reads" the page out loud

Media Features

- Media feature uses characteristics to differentiates one device from another device.
- Following is the list of media features.
  - width of the viewport.
  - height of the viewport.
  - orientation
    - Is the tablet/phone in landscape or portrait mode?
  - aspect-ratio
    - The aspect-ratio feature is specified as a ratio value representing the width-to-height ratio of the viewport.
  - resolution
    - The resolution feature is specified as a <resolution> value representing the pixel density of the output device. It is measures in dpi (dots per inch).
    - ……

Media Query Logical Operator

- The logical operators ***not***, ***and***, and ***only*** can be used to compose a complex media query.
- You can also combine multiple media queries into a single rule by separating them with commas.
  - ***and***
    - The ***and*** operator is used for combining multiple media features together into a single media query, requiring each chained feature to return `true` in order for the query to be true.
    - It is also used for joining media features with media types.

Devices with width within a range

```
@media (min-width: 768px) and (max-width: 991px) {…}
```

```
@media screen and (min-width: 400px) and (max-width: 700px) { … }
```

not/only logical Operators

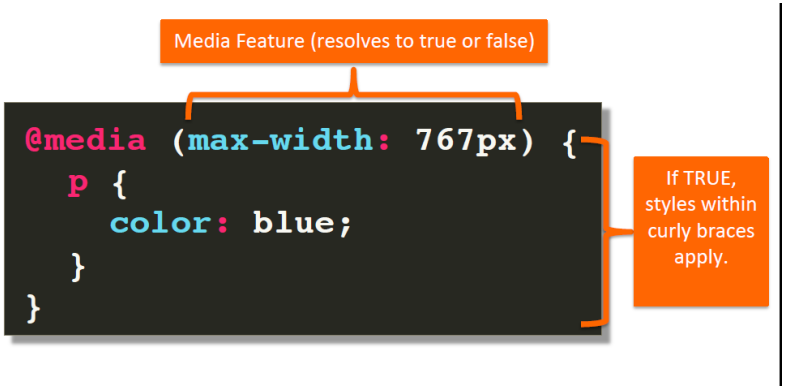
- ***not***
  - The `not` operator is used to negate a media query, returning `true` if the query would otherwise return `false`.
  - If you use the `not` operator, you must also specify a media type.
    - `@media not (screen and (max-width: 600px)) {……}`
- ***only***
  - The `only` operator is used to apply a style only if an entire query matches.
  - If you use the `only` operator, you must also specify a media type.
  - It is useful for preventing older browsers from applying selected styles.
    - When not using `only`, older browsers would interpret the query `screen and (max-width: 500px)` simply as `screen`, ignoring the remainder of the query, and applying its styles on all screens.
    - `@media only (screen and (max-width: 500px)) {……}`

comma operator

- ***, (comma)***
  - Commas are used to combine multiple media queries into a single rule.
  - Each query in a comma-separated list is treated separately from the others.
  - Thus, if any of the queries in a list is true, the entire media statement returns `true`.
  - In other words, lists behave like a logical ***or*** operator.
  - For instance, the following rule will apply its styles if the user's device has either a minimum height of 680px *or* is a screen device in portrait mode:
    - `@media (min-height: 680px), screen and (orientation: portrait) { ... }`

Media Feature (Min and Max Width)

- Most properties can be tested for a minimum or maximum value using the min- and max- prefixes, respectively.
  - The min prefix indicates a value of greater than or equal to while the max prefix indicates a value of less than or equal to.
  - For example,
    - max-width: 767px tests whether the display is in the range 0 to 767px.
    - min-width: 800px tests whether the display is at least 800 pixels wide.




```
@media (max-width: 800px) {…}
```

```
@media (min-width: 800px) {…}
```

Overlapping?

- `@media (min-width: 400px) {…}`
- `@media (min-width: 600px) {…}`
- `@media (min-width: 800px) {…}`

```
@media (min-width: 1200px) {
…
}

@media (min-width: 992px) and (max-width: 1199px) {
…
}
…
```

Careful not to overlap range boundaries!

Media Features (Orientation Syntax)

- The ‘orientation’ media feature is ‘portrait’ when the value of the ‘height’ media feature is greater than or equal to the value of the ‘width’ media feature. Otherwise ‘orientation’ is ‘landscape’.

- `@media all and (orientation:portrait) { … }`
- `@media all and (orientation:landscape) { … }`

- ```
  @media only screen and (orientation:
  landscape) {
    body {
      background-color: lightblue;
    }
  }
  ```
  ```

Media Features (aspect-ratio Syntax)

- The aspect-ratio feature is specified as a `<ratio>` value representing the width-to-height aspect ratio of the viewport.

```
/* Minimum aspect ratio */
@media (min-aspect-ratio: 8/5) {
  div {background: #9af; /* blue */  }
}

/* Maximum aspect ratio */
@media (max-aspect-ratio: 3/2) {
  div {background: #9ff;  /* cyan */  }
}
```

Complex Media Queries

- `@media only (screen and (min-width: 100px)),`
- `not (screen and (min-width: 100px)),`
- `not (print and (min-height: 100px)),`
- `(min-height: 100px) and (max-height: 1000px),`
- `(orientation: landscape)`
  - `{ p`
    - `{ background: red; }`
  - `}`

Identifying Breakpoints

- The point at which the media query delivers a new set of styles is known as a breakpoint.
- Some sites have just two layouts triggered at a single breakpoint. More commonly, responsive sites use three designs targeted at typical phone, tablet and desktop widths.
- How many you choose depends on the nature of your site's design.



Typical Device Breakpoints

- It is hard to create an exact breakpoint for each device. To keep things simple you could target five groups:

Example

```
/* Extra small devices (phones, 600px and down) */
@media only screen and (max-width: 600px) {...}

/* Small devices (portrait tablets and large phones, 600px and up) */
@media only screen and (min-width: 600px) {...}

/* Medium devices (landscape tablets, 768px and up) */
@media only screen and (min-width: 768px) {...}

/* Large devices (laptops/desktops, 992px and up) */
@media only screen and (min-width: 992px) {...}

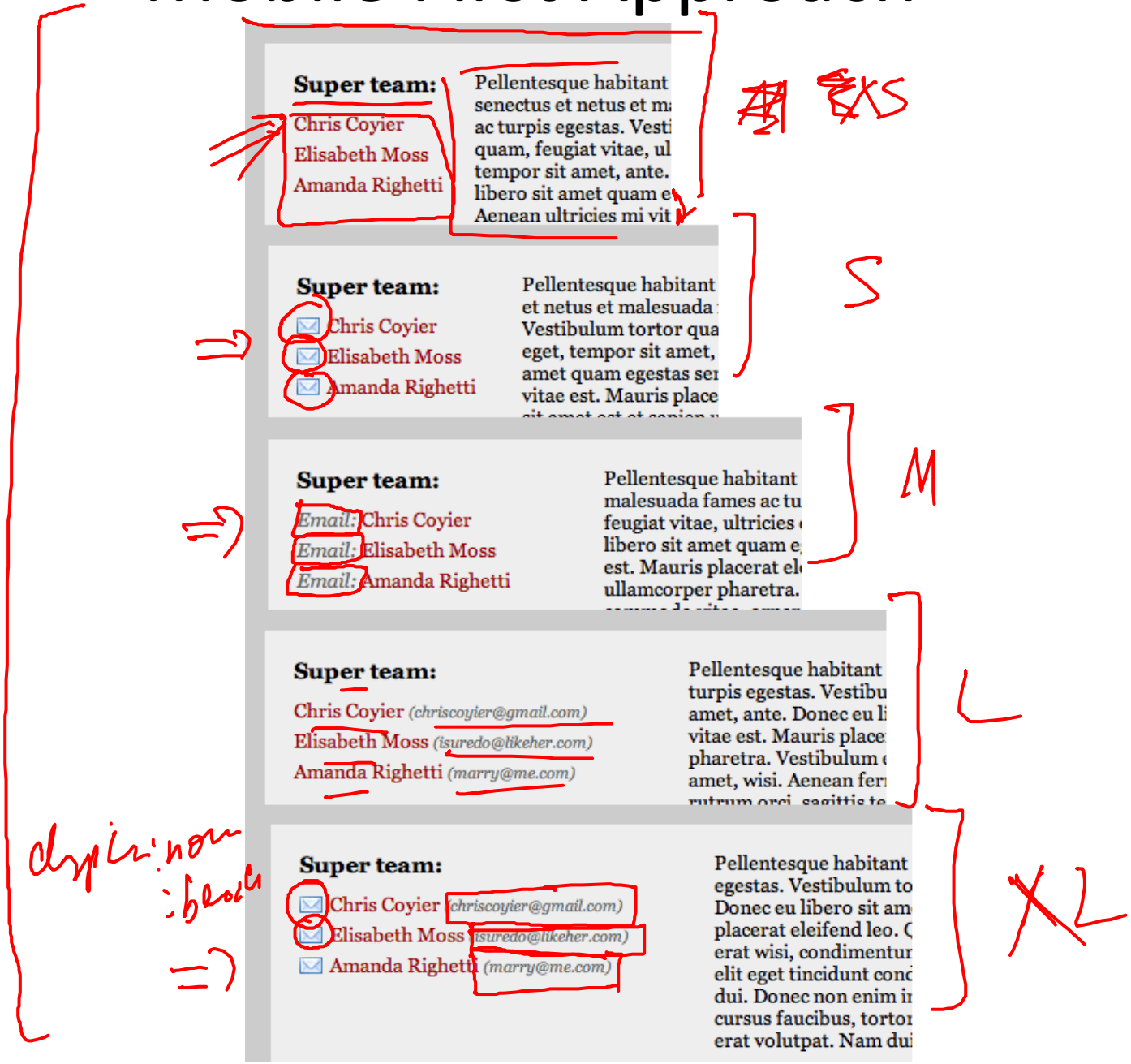
/* Extra large devices (large laptops and desktops, 1200px and up) */
@media only screen and (min-width: 1200px) {...}
```

Mobile First Approach

- One popular technique with using media queries is called *mobile first*.
- The mobile first approach includes using styles targeted at smaller viewports as the default styles for a website, then use media queries to add styles as the viewport grows.
- The operating belief behind mobile first design is that a user on a mobile device, commonly using a smaller viewport, should't have to load the styles for a desktop computer only to have them overwritten with mobile styles later.

```
/* Default styles first then media queries */
@media (min-width: 400px) {...}
@media (min-width: 600px) {...}
@media (min-width: 1000px) {...}
@media (min-width: 1400px) {...}
```

Mobile First Approach



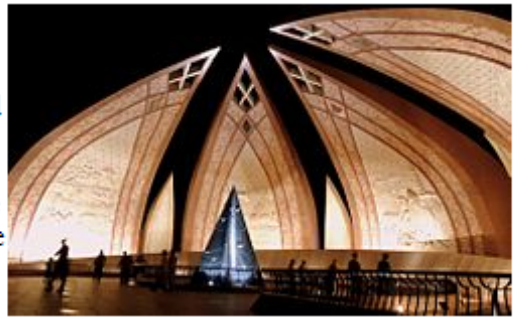
Small Viewport

1350 × 150

- Home
- News
- Contact
- About

- Islamabad
- Lahore
- Abbotabad
- Sahiwal
- Vehari
- Attock
- Wah

Islamabad



Pakistan Monument

Islamabad (/ɪzˈlɑːməˌbɑːd/; Urdu: اسلام آباد, Islāmābād, Urdu pronunciation: [ɪsˌlɑːməˈbɑːɖ]) is the capital city of Pakistan located within the federal Islamabad Capital Territory. With a population of two million, it is the 10th largest city of Pakistan, while the larger Islamabad Rawalpindi metropolitan area is the third largest in Pakistan with a population exceeding five million.[3][4][5] The city is the political seat of Pakistan and is administered by the Islamabad Metropolitan Corporation, supported by the Capital Development Authority (CDA).Islamabad is located in the Pothohar Plateau in the northeastern part of the country, between Rawalpindi District and the Margalla Hills National Park to the north. The region has historically been a part of the crossroads of Punjab and Khyber Pakhtunkhwa with the Margalla Pass acting as the gateway between the two regions.[6] Islamabad was built during the 1960s to replace Karachi

Medium Viewport

1350 × 150

Home News Contact About

- Islamabad
- Lahore
- Abbotabad
- Sahiwal
- Vehari
- Attock
- Wah

Islamabad

Islamabad (/ɪzˈlɑːməˌbɑːd/; Urdu: اسلام آباد, Islāmābād, Urdu pronunciation: [ɪsˌlɑːmɑ ˈbɑːɖ]) is the capital city of Pakistan located within the federal Islamabad Capital Territory. With a population of two million, it is the 10th largest city of Pakistan, while the larger Islamabad Rawalpindi metropolitan area is the third largest in Pakistan with a population exceeding five million.[3][4][5] The city is the political seat of Pakistan and is administered by the Islamabad Metropolitan Corporation, supported by the Capital Development Authority (CDA).Islamabad is located in the Pothohar Plateau in the northeastern part of the country, between Rawalpindi District and the Margalla



Big Viewport

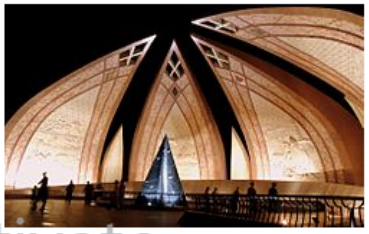
1350 × 150

Home News Contact About

- Islamabad
- Lahore
- Abbotabad
- Sahiwal
- Vehari
- Attock
- Wah

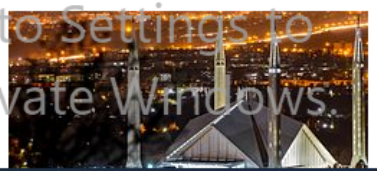
Islamabad

Islamabad (/ɪzˈlɑːməˌbɑːd/; Urdu: اسلام آباد, Islāmābād, Urdu pronunciation: [ɪsˌlɑːməˈbɑːɖ]) is the capital city of Pakistan located within the federal Islamabad Capital Territory. With a population of two million, it is the 10th largest city of Pakistan, while the larger Islamabad Rawalpindi metropolitan area is the third largest in Pakistan with a population exceeding five million.[3][4][5] The city is the political seat of Pakistan and is administered by the Islamabad Metropolitan Corporation, supported by the Capital Development Authority (CDA).Islamabad is located in the Pothohar Plateau in the northeastern part of the country, between Rawalpindi District and the Margalla Hills National Park to the north. The region has historically been a part of the crossroads of Punjab and Khyber Pakhtunkhwa with the Margalla Pass acting as the gateway between the two regions.[6] Islamabad was built during the 1960s to replace Karachi as Pakistan's capital. The city's master-plan divides the city into eight zones, including administrative, diplomatic enclave, residential areas, educational sectors, industrial sectors, commercial areas, and rural and green areas. The city is known for the presence of several parks and forests, including the Margalla Hills National Park and Shakarparian Park.[7] The city is home several landmarks, including the Faisal Mosque, the largest mosque in South Asia[8] and the fourth largest in the world. Other landmarks include the Pakistan's National Monument and Democracy Square.[9][10][11] Islamabad was built during the 1960s to replace Karachi as Pakistan's capital. The city's master-plan divides the city into eight zones, including administrative, diplomatic enclave, residential areas, educational sectors, industrial sectors, commercial areas, and rural and green areas. The city is known for the presence of several parks and forests, including the Margalla Hills National Park and Shakarparian Park.[7] The city is home several landmarks, including the Faisal Mosque, the largest mosque in South Asia[8] and the fourth largest in the world. Other landmarks include the Pakistan's National Monument and Democracy Square.[9][10][11]



Pakistan Monument

Islamabad (/ɪzˈlɑːməˌbɑːd/; Urdu: اسلام آباد, Islāmābād, Urdu pronunciation: [ɪsˌlɑːməˈbɑːɖ]) is the capital city of Pakistan located within the federal Islamabad Capital Territory. With a population of two million, it is the 10th largest city of Pakistan, while the larger Islamabad Rawalpindi metropolitan area is the third largest in Pakistan with a population exceeding five million.[3][4][5] The city is the political seat of Pakistan and is administered by the Islamabad Metropolitan Corporation, supported by the Capital Development Authority (CDA).Islamabad is located in the Pothohar Plateau in the northeastern part of the country, between Rawalpindi District and the Margalla Hills National Park to the north. The region has historically been a part of the crossroads of Punjab and Khyber Pakhtunkhwa with the Margalla Pass acting as the



Activate Windows
Go to Settings to activate Windows.

Linking Stylesheets

- Multiple CSS stylesheets can be linked within HTML. There are two main approaches:
  - `@import CSS rule.`
  - `<link> element.`

@import CSS Rule

- The `@import` CSS rule is used to import style rules from other style sheets. These rules must precede all other types of rules.
  - `@import url("style1.css")`
  - `@import url("style2.css")`
  - ……
- When importing multiple CSS files, the styles are applied in the order that they are imported in the HTML document.
- In the above code, all the rules from `style1.css` would be applied first, then the rules from `style2.css` will be applied. This can cause similar rules to overwrite the rules from the previously imported files.

@import CSS Rule

```
<!DOCTYPE html>
<html>
<head>
<style>

@import "resetStyles.css";
@import "BigScreen.css";
@import "mediumScreen.css";
@import "smallScreen.css";

</style>
</head>
<body>

<header>
    <img src="header.jpg" alt="Data" />
</header>
<nav>
    <ul class="topmenu">
      <li><a href="#home" class="active">Home</a></li>
      <li><a href="#news">News</a></li>
      <li><a href="#contact">Contact</a></li>
      <li><a href="#about">About</a></li>
    </ul>
```

`<link>` element

- `<link>` element can be used to link multiple stylesheets in a single HTML file.
- For each stylesheet you link to a page, you would just need to add an additional `<link>` element.
  - `<link href="style1.css" rel="stylesheet">`
  - `<link href="style2.css" rel="stylesheet">`
  - `...`
- When linking multiple CSS files, the styles are applied in the order that they are linked in the HTML document.
- In the above code, all the rules from `style1.css` would be applied first, then the rules from `style2.css` will be applied. This can cause similar rules to overwrite the rules from the previously linked files.

<link> element Example

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4
5  <link rel="stylesheet" type="text/css" href="resetStyles.css">
6  <link rel="stylesheet" type="text/css" href="BigScreen.css">
7  <link rel="stylesheet" type="text/css" href="mediumScreen.css">
8  <link rel="stylesheet" type="text/css" href="smallScreen.css">
9
10 <style>
11
12 //@import "resetStyles.css";
13 //@import "BigScreen.css";
14 //@import "mediumScreen.css";
15 //@import "smallScreen.css";
16
17 </style>
18 </head>
19 <body>
20
21 <header>
22     <img src="header.jpg" alt="Data" />
23 </header>
```

Activat
Windo
Go to Se
activate

Summary of Today's Lecture

- Fixed vs Fluid layouts
- Responsive Web Design
- Media Queries
  - Media Types
  - Media Features
  - Logical Operators
- Identifying Breakpoints
- Mobile First Approach
- Adding Media Queries
- Responsive Web Design Example

Resources

- Responsive Web Design by Ethan Marcotte
- https://www.w3.org/TR/css3-mediaqueries/
- https://abookapart.com/products/responsive-web-design
- http://learn.shayhowe.com/advanced-html-css/responsive-web-design/
- https://www.w3schools.com/cssref/pr_import_rule.asp
- https://developer.mozilla.org/en-US/docs/Web/CSS/@import
- https://discuss.codecademy.com/t/can-you-apply-multiple-stylesheets-to-a-single-page/369792